Mapping Mountain Land Cover: Objective and Methodology

Objective

Assess whether adding elevation and slope to Sentinel-2 imagery improves the distinction between scree, exposed bedrock and grassland in Posets–Maladeta, Spanish Pyrenees.

Input data

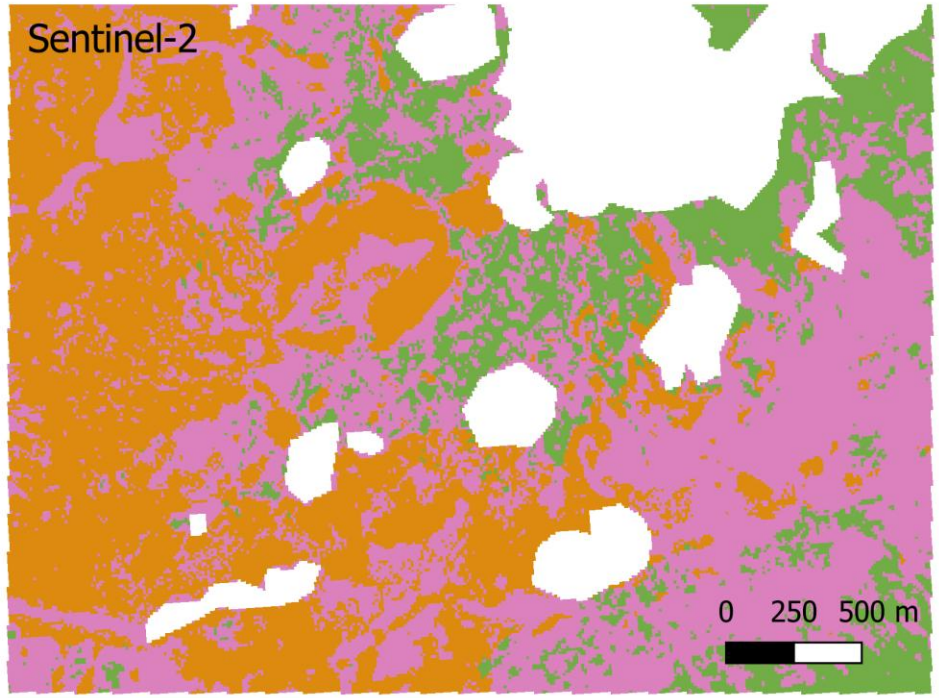
- **Sentinel-2 L2A:** 22 August 2024; six spectral bands and NDVI.
- **CNIG MDT02:** 2 m digital terrain model, aggregated to the Sentinel-2 grid; elevation and slope used as predictors.
- **PNOA orthophotography:** 2024, 25 cm resolution; reference for training-sample selection and independent image interpretation.

Methodology

1. **Prepare the data:** Align predictors to a common grid of approximately 10 m and mask manually identified exclusions and missing data.
2. **Define training simples:** Select 13 reference polygons representing the three classes, providing 1,914 training pixels.
3. **Compare two Random Forest models:** Use spectral predictors alone, then add elevation and slope, with the same model settings and validation folds.
4. **Evaluate and map:** Apply four-fold validation by whole training polygon, generate both maps, and assess 60 random cells stratified by the terrain model's predicted classes, outside the training polygons.

Exploratory land-cover classification.

Exploratory Mapping of Scree, Exposed Bedrock and Grassland in Posets–Maladeta



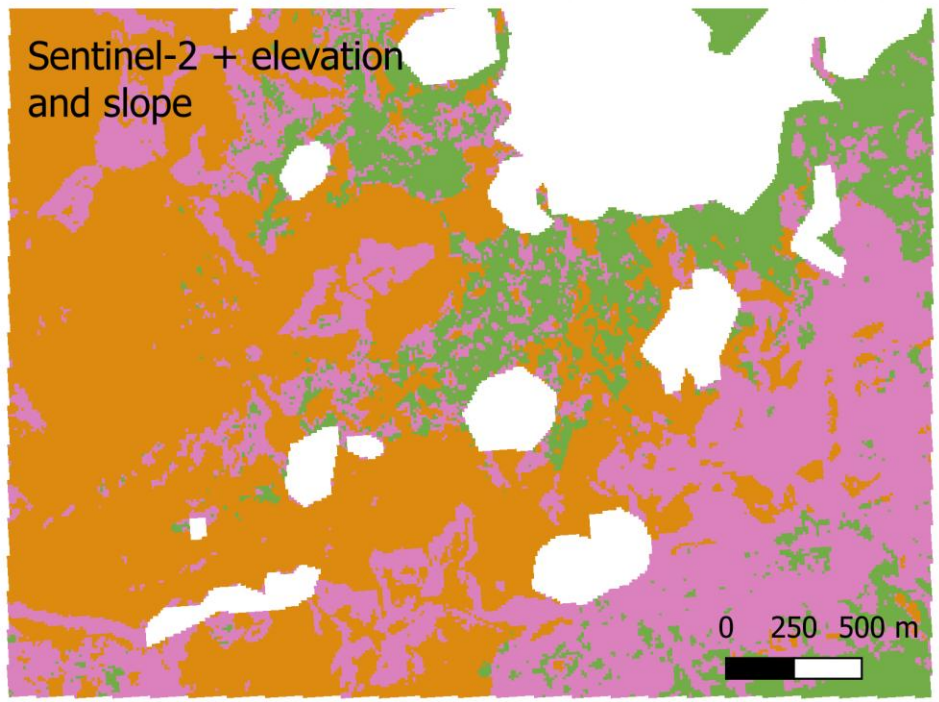
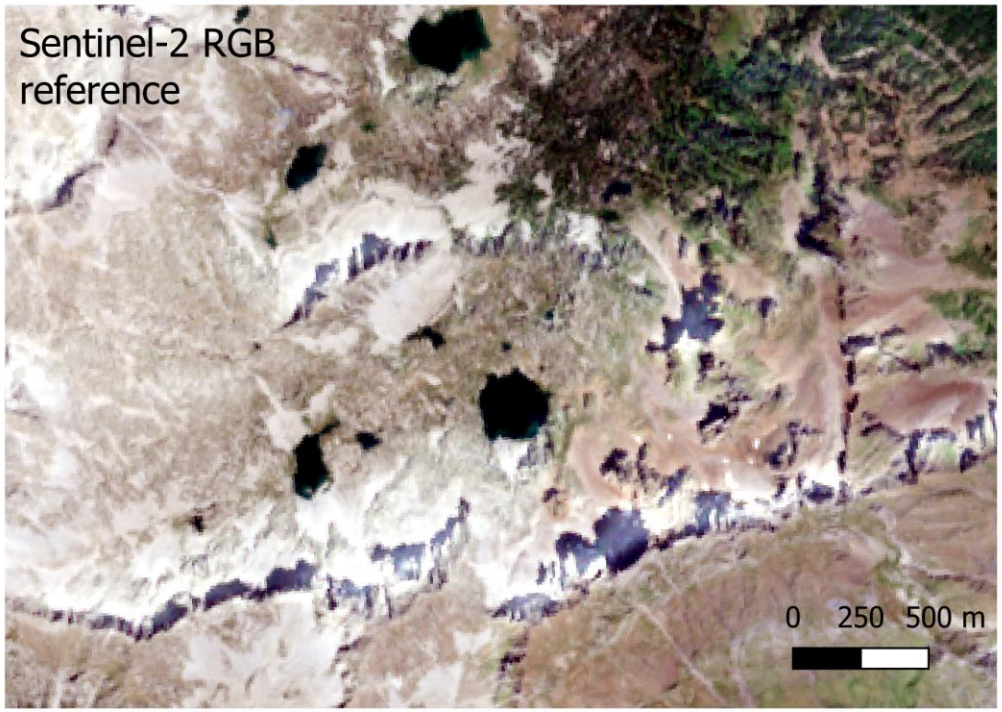


Image classification
- Scree
- Exposed bedrock
- Grassland
- Excluded / missing data



Sentinel-2 imagery: 22 August 2024. Terrain data: CNIG MDT02. Grid spacing: approximately 10 m; bands B11 and B12 resampled from 20 m. Excluded areas and pixels with missing data remain unclassified. Classes represent land cover,

Model Performance and Validation

Polygon-based cross-validation

Four folds, keeping all pixels from each training polygon in the same fold. 1,914 pixels from 13 polygons.

Metric	Sentinel-2	Sentinel-2 + terrain
Overall accuracy	89.6%	97.0%
Balanced accuracy	88.3%	95.6%
Macro F1-score	0.865	0.957

Independent map assessment

60 randomly selected cells, stratified by the terrain model's predicted classes and outside the training polygons. Reference labels assigned through PNOA interpretation.

Metric	Sentinel-2	Sentinel-2 + terrain
Correct predictions — 53 labelled cells	34/53	36/53
Weighted accuracy — 53 labelled cells	60.5%	67.6%

Accuracy weighted by sampling-stratum size, conditional on the 53 cells assigned to the three target classes. Six unresolved mixed cells and one lake are excluded. References are based on PNOA interpretation, without field verification.

Terrain increased weighted accuracy by 7.1 percentage points among the 53 labelled cells. Scree–bedrock confusion remained the main limitation. Lower performance in the independent assessment highlights the need for more representative training samples and further validation.